AFRICAN STANDARD

ARS 464

Third Edition 2022

Milled rice — Specification

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Table of contents

1 Scope 1
2 Normative references 1
3 Definitions 2
4 Quality requirements 6
4.2 General requirements 6
4.3 Specific requirements 6
4.3.1 Classification 6
4.3.2 Grading 7
5 Contaminants 7
5.1 Metals 7
5.2 Pesticide residues 8
6 Hygiene 8
7 Packaging 8
8 Labelling 8
9 Sampling methods 9
Annex A (normative) Methods of analysis for rice specifications ..... **Error! Bookmark not defined.**
Annex B (normative) Determination of waxy rice in parboiled rice ..... **Error! Bookmark not defined.**
Bibliography 10

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Foreword

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This African Standard was prepared by the ARSO Technical Committee on Cereals, pulses and derived products (ARSO TC 12).

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Introduction

Rice is a widely consumed cereal grain in Africa grown, from river deltas to mountainous regions and mainly uses rainfed systems. Predicted demands for rice remain strong. In Africa, where rice is the most rapidly growing food source, about 30 million tons more rice will be needed by 2035, representing an increase of 130 % in rice consumption from 2010.
This standard does not apply to parboiled rice.

This standard has been revised to take into account:

a) the needs of the market for the product;

b) the need to facilitate fair domestic, regional and international trade and prevent technical barriers to trade by establishing a common trading language for buyers and sellers;

c) the structure of the CODEX, UNECE, USA, ISO and other internationally significant standards;

d) the needs of the producers in gaining knowledge of market standards, conformity assessment, commercial cultivars and crop production process;

e) the need to transport the product in a manner that ensures keeping of quality until it reaches the consumer;

f) the need for the plant protection authority to certify, through a simplified form, that the product is fit for cross-border and international trade without carrying plant disease vectors;

g) the need to promote good agricultural practices that will enhance wider market access, involvement of small-scale traders and hence making farming a viable means of wealth creation; and

h) the need to ensure a reliable production base of consistent and safe crops that meet customer requirements.

This African Standard is a technical revision of the earlier ARS 464:2013(E), *Rice — Specification* which is hereby superseded and cancelled.

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Milled rice — Specification

1 Scope

This African Standard specifies the requirements and methods of sampling and test for milled rice of the varieties grown from *Oryza spp*. intended for human consumption.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ARS 53, *General principles of food hygiene — Code of practice*

ARS 56, *Prepackaged foods — Labelling*

ARS 858, *Rough (paddy) rice — Specification*

ARS 859, *Brown rice — Specification*

CODEX STAN 193, *Codex general standard for contaminants and toxins in food and feed*

ISO 605*, Pulses — Determination of impurities, size, foreign odours, insects, and species and variety — Test methods*

ISO 712*, Cereals and cereal products — Determination of moisture content — Routine reference method*

ISO 5223, *Test sieves for cereals*

ISO 6561-1, *Fruits, vegetables and derived products — Determination of cadmium content — Part 1: Method using graphite furnace atomic absorption spectrometry*

ISO 6561-2, *Fruits, vegetables and derived products — Determination of cadmium content — Part 2: Method using flame atomic absorption spectrometry*

ISO 6633, *Fruits, vegetables and derived products — Determination of lead content — Flameless atomic absorption spectrometric method*

ISO 6888-1, *Microbiology of food and animal feeding stuffs — Horizontal method for the enumeration of coagulase-positive staphylococci (Staphylococcus aureus and other species) — Part 1: Technique using Baird-Parker agar medium*

ISO 16050, *Foodstuffs — Determination of aflatoxin B_1, and the total content of aflatoxin B_1, B_2, G_1 and G_2 in cereals, nuts and derived products — High performance liquid chromatographic method*

ISO 6579-1, *Microbiology of the food chain* — Horizontal method for the detection, enumeration and serotyping of Salmonella — *Detection of Salmonella spp*

ISO 21527-2, *Microbiology of food and animal feeding stuffs — Horizontal method for the enumeration of yeasts and moulds — Part 2: Colony count technique in products with water activity less than or equal to 0.95*

ISO 24333, *Cereals and cereal products — Sampling*

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ISO 27085, *Animal feeding stuffs — Determination of calcium, sodium, phosphorus, magnesium, potassium, iron, zinc, copper, manganese, cobalt, molybdenum, arsenic, lead and cadmium by ICP-AES*

AOAC Official Method 2001.04, *Determination of Fumonisins B_1 and B_2 in corn and corn flakes — Liquid chromatography with immunoaffinity column cleanup*

3 Terms and Definitions

For the purpose of this African Standard, the following terms and definitions apply.

3.1
paddy
paddy rice
rough rice
whole or broken kernels of rice from (*Oryza glaberrima, Oryza sativa, Oryza longistaminata*) retaining its husk after threshing

3.2
husked rice
brown rice
cargo rice
paddy from which the husk only has been removed

NOTE The processes of husking and handling may result in some loss of bran.

3.3
milled rice
whole or broken kernels of rice (*Oryza* spp) from which the germ, embryo or at least the outer bran layer have been removed,

3.3.1
undermilled rice
milled rice obtained by milling husked rice, but not to the degree necessary to meet the requirements of well-milled rice

3.3.2
well-milled rice
milled rice obtained by milling husked rice in such a way that most of the bran and part of the embryo have been removed

3.3.3
extra well-milled rice
milled rice obtained by milling husked rice in such a way that almost all of the bran and the embryo have been removed

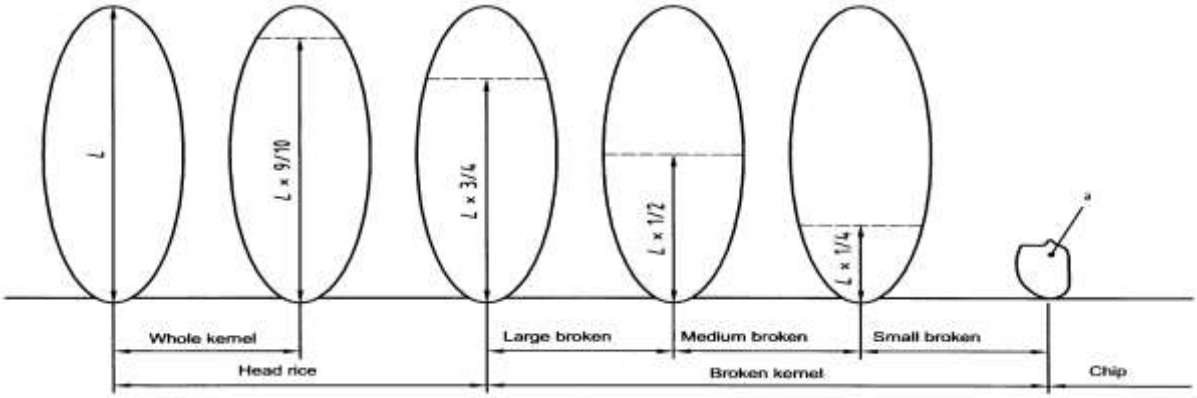
3.4
waxy rice
glutinous rice
varieties of rice whose kernels have a white and opaque appearance

NOTE The starch of waxy rice consists almost entirely of amylopectin. The kernels have a tendency to stick together after cooking.

3.5
whole kernel
husked or milled kernel without any broken part, or part of kernel with a length greater than or equal to nine-tenths of the average length of the test sample kernels

NOTE See Figure 1.

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[a] Not passing through a round perforation of 1.4 mm in diameter

L is the average length

Figure 1 — Size of kernels, broken kernels and chips

3.6
head rice
whole kernel or part of kernel with a length greater than or equal to three-quarters of the average length of the test sample kernels

NOTE See Figure 1.

3.7
broken kernels
pieces of rice that are less than three-quarters of a whole kernel and includes grains of rice in which part of the endosperm is exposed or rice without a germ. If the piece is more than three-quarters of a kernel, it is considered whole

3.7.1
large broken kernel
part of kernel with a length less than three-quarters but greater than one-half of the average length of the test sample kernels

NOTE See Figure 1.

3.7.2
medium broken kernel
part of kernel with a length less than or equal to one-half but greater than one-quarter of the average length of the test sample kernels

NOTE See Figure 1.

3.7.3
small broken kernel
part of kernel with a length less than or equal to one-quarter of the average length of the test sample kernels but which does not pass through test sieve with round apertures having diameter 1.4 mm

NOTE See Figure 1.

3.8
chip
part of kernel which passes through a test sieve complying with ISO 5223, and with round apertures having diameter 1.4 mm

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3.9
average length, *L*
arithmetic mean of the length of the test sample kernels that are not immature or malformed and without any broken parts

3.10
foreign matter
all organic and inorganic material (such as plant parts, sand, soil, glass) other than rice

3.10.1
inorganic foreign matter
components, such as stone, sand and dust

3.10.2
organic foreign matter
any animal or plant matter (seed coats, straws, weeds) other than grain of maize and damaged maize grain.

3.10.3
filth
impurities of animal origin such as eggs, larvae, dead insects and their fragments

3.11
heat-damaged kernel
head rice or broken kernel that has changed its normal colour as a result of microbiological heating

NOTE This category includes kernel that is yellow to dark yellow in the case of non-parboiled rice and orange to dark orange in the case of parboiled rice, due to a microbiological alteration.

3.12
damaged kernel
head rice or broken kernel showing evident deterioration due to moisture, pests, disease or other causes, but excluding heat-damaged kernels

3.12.1
spotted kernel
whole or broken kernel showing a well defined small circle of dark colour or more or less regular shape

3.12.2
stained kernel
whole or broken kernel which has undergone on a small area of its surface an obvious change in its natural colour. The stains maybe of different colours e.g., blackish, reddish and brown. Deep black striations are also considered stains.

3.12.3
specks
head rice or broken kernel of parboiled rice of which more than one-quarter of the surface is dark brown or black in colour due to the parboiling process

3.12.4
shrivel kernel
kernel which has become shrunken and wrinkled from great heat or lack of moisture

3.12.5
black kernel
kernel showing a distinctly dark colouration

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3.12.6
over-dried damaged
defective grains caused by overheating during artificial drying. It can be detected where grain is hot, exhibits an unusual odour, exhibits significant sprouting (greater than 10%) or other evidence of weather damage

3.12.7
smutty rough rice
rough rice which contains more than 3.0 percent of smutty kernels

3.13
chalky kernel
head rice or broken kernel of non-parboiled rice, except waxy rice, whose whole surface has an opaque and floury appearance

3.14
red kernel
head rice or broken kernel having a red bran covering more than one-quarter of its surface

3.15
red-streaked kernel
head rice or broken kernel with red bran streaks of length greater than or equal to one-half of the average length, but where the surface covered by these red streaks is less than one-quarter of the total surface

3.16
milling yield
estimate of the quantity of whole kernels and total milled rice (whole and broken kernels combined) that are produced in the milling of rough rice to a well-milled degree

3.17
poisonous, toxic and/or harmful seeds
any seed which if present in quantities above permissible limit may have damaging or dangerous effect on health, organoleptic properties or technological performance such as Jimson weed — dhatura (*D. fastuosa* Linn and *D. stramonium* Linn.) corn cockle (*Agrostemma githago* L., *Machai Lallium remulenum* Linn.) Akra (Vicia species), *Argemone mexicana*, Khesari and other seeds that are commonly recognized as harmful to health

3.18
green/immature kernel
whole or broken kernel, which is undeveloped and may be green in colour

3.18.1
yellow kernel
whole kernel, which has undergone, totally or partially, through heating or other causes, a change in its natural colour and has taken a lemon or orange-yellow tone

3.18.2
amber kernel
whole kernel, which has undergone thorough heating or other causes, a slight uniform change in colour over the whole surface; this change alters the colour of the kernel to a slight amber-yellow

3.19
food grade packaging material
packaging material, made of substances which are safe and suitable for their intended use and which will not impart any toxic substance or undesirable odour or flavour to the product

3.20
parboiled rice
milled rice in which the starch has been gelatinized by soaking, steaming, and drying

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4 Requirements

4.1 Raw materials

Milled rice shall be obtained from rough or brown rice of sound quality, free from sand, having characteristic odour and flavour and complying with ARS 858 or ARS 859.

4.2 General requirements

Milled rice shall meet the following general requirements/limits:

(a) clean, wholesome, uniform in size, colour and shape;

(b) safe and suitable for human consumption;

(c) free from abnormal flavour, musty, or other undesirable odour and objectionable smell;

(d) free from live pests;

(e) free from micro-organisms and substances originating from micro-organisms, fungi or other poisonous or deleterious substances in amounts that may constitute a hazard to human health.

4.3 Specific requirements

4.3.1 Classification

Rice shall be classified as long grain, medium grain or short grain. The classification shall be done in accordance with one of the following specifications:

4.3.1.1 Option 1: Kernel length/width ratio

(i) **long** grain rice: milled rice with a length/width ratio of **3.0 or more**;

(ii) **medium** grain rice: milled rice with a length/width ratio of **2.0 – 2.9**

(iii) **short** grain rice: milled rice with a length/width ratio of **1.9 or less**.

4.3.1.2 Option 2: The kernel length

(i) **Long grain rice,** with a kernel length of 6.6 mm or more.

(ii) **Medium grain rice,** with a kernel length of 6.2 mm or more but less than 6.6 mm.

(iii) **Short grain rice** has a kernel length of less than 6.2 mm.

4.3.1.3 Option 3: Combination of the kernel length and the length/width ratio

4.3.1.3.1 Long grain rice has either:

(i) kernel length of more than 6.0 mm and with a length/width ratio of more than 2 but less than 3, or;

(ii) a kernel length of more than 6.0 mm and with a length/width ratio of 3 or more.

4.3.1.3.2 Medium grain rice with a kernel length of more than 5.2 mm but not more than 6.0 mm and a length/width ratio of less than 3.

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4.3.1.3.3 Short grain rice with a kernel length of 5.2 mm or less and a length/width ratio of less than 2.

4.3.2 Grading

Milled rice grains for human consumption shall be graded into three grades on the basis of the tolerable limits established in Table 1 which shall be additional to the general requirements set out in this standard.

Table 1 — Specific requirements

S/No.	Characteristics	Maximum limits			Test Method
		Grade 1	Grade 2	Grade 3	
(1)	Broken, %	5	15	25	ISO 605
(2)	Damaged rice, %	1.5	2	3.0	
(3)	Chalky %	2	4	10	
(4)	Red or red streaked, %	2	6	12	
(5)	Immature grains, %	1	1.5	2	
(6)	Other contrasting varieties, %	1	2	3	
(7)	Organic matter, %	0.1	0.2	0.5	
(8)	Inorganic matter, %	0.1	0.1	0.1	
(9)	Paddy grains, %	0.3	0.3	0.3	
(10)	Live insect infestation, kg	Nil	Nil	Nil	
(11)	Filth, m/m %	0.1	0.1	0.1	
(12)	Moisture contents, m/m %	14	14	14	ISO 712
(13)	Foreign matter, % m/m	0.2	0.3	0.6	ISO 605
(14)	Total Aflatoxin (AFB1+AFB2+AFG1 +AFG2)), ppb	10			ISO 16050
(15)	Aflatoxin B1 only, ppb	5			
(16)	Fumonisin B_1 and B_2 ppm	2			AOAC 2001.04

4.3.3 Ungraded milled rice

Ungraded milled rice shall be milled rice which does not fall within the requirements of Grades 1, 2, and 3 of this standard but is not rejected rice grains.

5 Contaminants

5.1 Heavy metal contaminants

Milled rice shall comply with those limits for heavy metals specified in CODEX STAN 193.

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5.2 Pesticide residues

Milled rice shall comply with those maximum pesticide residue limits established by the Codex Alimentarius Commission for this commodity

6 Hygiene

Milled rice shall be produced and handled under hygienic conditions in accordance with ARS 53.

7 Weights and measures

Milled rice shall be packaged in accordance with the weights and measures regulations of the destination country.

NOTE: Maximum package weight of 50 kg where human loading and offloading is involved'.

8 Packaging

Milled rice shall be packed in food grade packaging material, which will safeguard the hygienic, nutritional and organoleptic qualities of the products**.**

9 Labelling

The following specific labelling requirements shall apply and shall be legibly and indelibly marked in accordance with the requirements of ARS 56:

(i) product name as "Milled Rice";

(ii) variety;
- Long grain milled rice
- Medium grain milled rice
- Short grain milled rice
- Mixed milled rice

(iii) grade;

(iv) name, address and physical location of the manufacturer/ packer/importer;

(v) lot/batch/code number;

(vi) net weight, in kg;

(vii) the declaration "Food for Human Consumption''

(viii) storage instruction as "Store in a cool and dry place away from any contaminants'';

(ix) crop year;

(x) packing date;

(xi) best before date

(xii) instructions on disposal of used package;

(xiii) country of origin;

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(xiv) a declaration on whether the milled rice was genetically modified or not.

10 Sampling

Sampling shall be done in accordance with the ISO 24333.

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Bibliography

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United States Standards for Milled Rice, Updated July 2005

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